

Open Source Breathes Life into Robots, Thermostats, and even E-Waste!

The wide range of devices being built using open platforms and development boards is phenomenal. From a cute robot that can serve coffee to a project that aims to convert e-waste into educational toys, there are a lot of open source-based innovations happening around the world. Some of these are for kicks, while others improve the lives of many; but all are inspiring, as we see month after month. So, here is another set of awesome innovations to mull over and maybe even contribute to.

Smart thermostat for the Internet of Things era

<http://www.tado.com/>



Tado is a smart Internet-connected thermostat that eliminates knobs, buttons and even displays from the thermostat structure—it is simply a smooth, smart box that connects with a smartphone application. It connects to your heating system using its old thermostat or through the wiring itself, and enables you to control it through the intuitive smartphone app. The name Tado is derived from the Japanese greetings *tadaima* and *okaeri*, which mean "I'm back home" and "Welcome home." True to its name, Tado knows when someone enters or leaves a room, and automatically turns the A/C on or off, accordingly. By connecting to the Internet, Tado adjusts the indoor climate based on weather forecasts, user behaviour, and the individual building's characteristics. According to a pilot study by the company, Tado is able to save up to 30 per cent on air conditioning related power bills.

The Open Twist: Tado connects to the Internet using Thingsquare Mist. The Thingsquare platform, developed by Adam Dunkels of Contiki fame, basically gives everyday objects a unique identity and connects them to smartphone apps, so that they become part of the Internet of Things (IoT). A device maker just needs to add a programmable wireless chip that runs the Contiki-based, IPv6-compatible,

open source firmware called Thingsquare Mist. The wireless chip and the Thingsquare firmware securely connect the device to the Thingsquare cloud backend server. The cloud backend provides an API for smartphone apps. Thingsquare works with a range of wireless chips from several different chip manufacturers, and also supports different types of wireless radios. With the cloud-based, easy-to-use development platform, developers can come up with a range of applications to put the 'connected' device to good use! In a company press release, Tado co-founder and CTO Johannes Schwarz claims, "By choosing Thingsquare Mist, we were able to develop an Internet-connected product within a surprisingly short period of time. Internet-connectivity is at the heart of our product and our customers value hassle-free installation, something that Thingsquare Mist helped us bring to our system."

Automate your environment, very easily

<http://www.wigwag.com/>

WigWag is a simple solution that helps you build intelligent environments without any complicated networking or programming. You just need to write the rules in a simple 'when x happens, then do y' format, using an app that works on smartphones and tablets, and WigWag will do the rest. For instance, you can set up a security system that notifies you when there is movement outside your door by pairing a sensor unit with the app, and saying 'when motion; then notify phone.' You can set up any number of rules, to automate any number of devices, at any number of locations (even remote). You can also share your rules with friends who are visiting for the weekend, so they also feel at home with the automated devices in your house—or they can alter the rules to suit their own convenience!

The WigWag solution comprises three devices: the sensor, relay and Glowline. The WigWag relay connects WigWag and third-party devices to the company's cloud service, which allows the connected devices to interact